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DEEP LEARNING

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NLP / GENERATIVE AI

Interview Questions and Answers

General

1. What is machine learning? What are the different machine learning methods? (<https://aiml.com/what-is-machine-learning/>)
2. Distinguish between Structured and Unstructured Data (<https://aiml.com/distinguish-between-structured-and-unstructured-data/>)

Deep Learning

3. What is Deep Learning? Discuss its key characteristics, working and applications (<https://aiml.com/what-is-deep-learning/>)
4. What are the advantages and disadvantages of Deep Learning? (<https://aiml.com/what-are-the-advantages-and-disadvantages-of-deep-learning/>)
5. How does Deep Learning methods compare with traditional Machine Learning methods? (<https://aiml.com/how-does-deep-learning-methods-compare-with-traditional-machine-learning-methods/>)

6. Explain the basic architecture of a Neural Network, model training and key hyper-parameters (<https://aiml.com/what-is-the-basic-architecture-of-an-artificial-neural-network-ann/>)
7. What is a Perceptron? What is the role of bias in a perceptron (or neuron)? (<https://aiml.com/what-is-a-perceptron/>)
8. What is a Multilayer Perceptron (MLP), also commonly known as Feed Forward Neural Network? (<https://aiml.com/what-is-a-multilayer-perceptron-mlp/>)
9. What do you mean by pretraining, finetuning and transfer learning? (<https://aiml.com/what-do-you-mean-by-pretraining-finetuning-and-transfer-learning-in-the-context-of-machine-learning-or-language-modeling/>)
10. What is an activation function, and what are the most common choices for activation functions? (<https://aiml.com/what-is-an-activation-function-what-are-the-different-types-of-activation-functions-discuss-their-pros-and-cons/>)
11. What are some options to address overfitting in Neural Networks? (<https://aiml.com/what-are-some-options-to-address-overfitting-in-neural-networks/>)
12. What is the vanishing and exploding gradient problem, and how are they typically addressed? (<https://aiml.com/what-do-you-mean-by-vanishing-and-exploding-gradient-problem-and-how-are-they-typically-addressed/>)
13. Compare the different Sequence models (RNN, LSTM, GRU, and Transformers) (<https://aiml.com/compare-the-different-sequence-models-rnn-lstm-gru-and-transformers/>)
14. What is Rectified Linear Unit (ReLU) activation function? Discuss its advantages and disadvantages (<https://aiml.com/what-is-rectified-linear-unit-relu-activation-function/>)
15. What is the “dead ReLU” problem and, why is it an issue in Neural Network training? (<https://aiml.com/what-is-dying-relu-or-dead-relu-and-why-is-this-a-problem-in-neural-network-training/>)

16. Briefly describe the architecture of a Recurrent Neural Network (RNN) (<https://aiml.com/briefly-describe-the-architecture-of-a-recurrent-neural-network-rnn/>)
17. What are the advantages and disadvantages of a Recurrent Neural Network (RNN)? (<https://aiml.com/what-are-the-advantages-and-disadvantages-of-a-recurrent-neural-network-rnn/>)
18. What is backpropagation? (<https://aiml.com/what-is-backpropagation/>)
19. How does dropout work? (<https://aiml.com/what-is-dropout/>)
20. What is Long-Short Term Memory (LSTM)? (<https://aiml.com/what-is-long-short-term-memory-lstm/>)
21. What are generative adversarial networks (GANs), and how are they used in deep learning?

Transformers

22. What are Transformers? Discuss the evolution and major breakthroughs in transformer models (<https://aiml.com/what-are-transformers/>)
23. Explain the Transformer Architecture (<https://aiml.com/explain-the-transformer-architecture/>)
24. What are the primary advantages of transformer models? (<https://aiml.com/what-are-the-main-advantages-of-the-transformer-models/>)
25. What are the limitations of transformer models? (<https://aiml.com/what-are-the-drawbacks-of-transformer-models/>)
26. Explain Self-Attention, and Masked Self-Attention as used in Transformers (<https://aiml.com/explain-self-attention-and-masked-self-attention-as-used-in-transformers/>)
27. What is Multi-head Attention and how does it improve model performance over single Attention head? (<https://aiml.com/what-is-multi-head-attention-and-how-does-it-improve-model-performance-over-single-attention-head/>)

28. Explain Cross-Attention and how is it different from Self-Attention?
(<https://aiml.com/explain-cross-attention-and-how-is-it-different-from-self-attention/>)

Natural Language Processing (NLP)

29. What is Natural Language Processing (NLP) ? List the different types of NLP tasks (<https://aiml.com/what-is-natural-language-processing-nlp-list-some-of-the-most-common-nlp-tasks/>)
30. What are some common applications of natural language processing (NLP)? (<https://aiml.com/what-are-some-of-the-most-common-applications-of-nlp/>)
31. What are Language Models? Discuss the evolution of Language Models over time (<https://aiml.com/what-are-language-models/>)
32. What is Bag-of-Words Model? Explain using example (<https://aiml.com/what-is-bag-of-words-model/>)
33. What are the advantages and disadvantages of Bag-of-Words model? (<https://aiml.com/what-are-the-advantages-and-disadvantages-of-bag-of-words-model/>)
34. What is topic modeling? Discuss its working, applications, and the pros and cons (<https://aiml.com/what-is-topic-modeling/>)
35. How is topic modeling used in text summarization? (<https://aiml.com/how-is-topic-modeling-used-in-text-summarization/>)
36. What is an n-gram model? (<https://aiml.com/what-is-an-n-gram-model/>)
37. What are word embeddings, and how are they used in NLP?
38. What are generative models, and how are they used in machine learning?

Supervised Learning

39. What is supervised learning? What are some common algorithms used in supervised learning (<https://aiml.com/what-is-supervised-learning/>)

40. Explain the concept of Linear Regression (<https://aiml.com/what-is-linear-regression/>)
41. What are the assumptions in a Linear Regression model?
(<https://aiml.com/what-are-the-assumptions-of-linear-regression/>)
42. What are the key evaluation criteria for Linear Regression model?
(<https://aiml.com/what-are-some-of-the-evaluation-criteria-used-to-assess-the-fit-of-a-linear-regression-model/>)
43. What is classification, and discuss the different types of classification?
What are some common classification algorithms? (<https://aiml.com/what-is-classification/>)
44. What is overfitting, and how can it be prevented in supervised learning?
(<https://aiml.com/how-can-overfitting-be-mitigated/>)
45. What is underfitting and how can it be prevented? (<https://aiml.com/what-is-underfitting/>)
46. What is Logistic Regression? Describe the process of how logistic regression is used to fit data (<https://aiml.com/what-is-logistic-regression/>)
47. What are the advantages and disadvantages of logistic regression?
(<https://aiml.com/what-are-advantages-and-disadvantages-of-logistic-regression/>)
48. What is a naive bayes classifier? Explain how does Naive Bayes work
(<https://aiml.com/how-does-naive-bayes-work/>)
49. What is the basic idea of Support Vector Machine (SVM) and Maximum Margin? (<https://aiml.com/what-is-the-basic-idea-of-support-vector-machine-svm-and-maximum-margin/>)
50. What are common choices to use for kernels in SVM?
(<https://aiml.com/what-are-common-choices-to-use-for-kernels-in-svm/>)
51. What is the kernel trick in SVM? (<https://aiml.com/what-is-the-kernel-trick-in-svm/>)
52. How do you evaluate the performance of a classification model? Discuss confusion matrix, precision, recall, F1-score in this context

(<https://aiml.com/how-would-you-evaluate-a-classification-based-model-using-a-confusion-matrix/>)

- 53. What is a ROC curve? (<https://aiml.com/how-would-you-evaluate-a-classification-model-using-roc-auc/>)
- 54. How can you handle imbalanced datasets in classification tasks? (<https://aiml.com/how-would-you-address-an-imbalanced-classification-problem/>)
- 55. What is the difference between a generative and a discriminative model? (<https://aiml.com/what-is-the-difference-between-discriminative-and-generative-models/>)
- 56. What does L1 regularization (Lasso) mean? (<https://aiml.com/what-does-l1-regularization-lasso-mean/>)
- 57. What does L2 regularization (Ridge) mean? (<https://aiml.com/what-does-l2-regularization-ridge-mean/>)

Ensemble Learning

- 58. What is a Decision Tree? (<https://aiml.com/what-is-a-decision-tree/>) What are the advantages and disadvantages of using a Decision Tree (<https://aiml.com/what-are-the-advantages-and-disadvantages-of-using-a-decision-tree/>)
- 59. (<https://aiml.com/what-is-bagging/>) What is Bagging? How do you perform bagging and what are its advantages? (<https://aiml.com/what-is-bagging/>)
- 60. What is Gradient Boosting? Describe how does the Gradient Boosting algorithm work (<https://aiml.com/what-is-gradient-boosting-gbm/>)
- 61. (<https://aiml.com/what-is-random-forest-2/>) Explain the concept and working of the Random Forest model (<https://aiml.com/what-is-random-forest-2/>)
- 62. What is XGBoost? How does it improve upon standard GBM? (<https://aiml.com/what-is-xgboost-and-how-does-it-improve-upon-standard-gbm/>)

63. What is the difference between Adaboost and Gradient boost?
(<https://aiml.com/what-is-the-difference-between-adaboost-vs-gradient-boost/>)
64. What is the difference between Decision Trees, Bagging, Boosting and Random Forest? (<https://aiml.com/what-is-the-difference-between-decision-trees-bagging-and-random-forest/>)
65. How is Gradient Boosting different from Random Forest?
(<https://aiml.com/how-is-gradient-boosting-different-from-random-forest/>)
66. GBM vs Random Forest: which algorithm should be used when?
(<https://aiml.com/in-what-situations-is-it-more-advisable-to-use-gbm-instead-of-random-forest-and-vice-versa/>)
67. Distinguish between a Weak learner and a Strong Learner
(<https://aiml.com/distinguish-between-a-weak-learner-vs-strong-learner/>)
68. What parameters can be tweaked for a Random Forest model? Explain in detail (<https://aiml.com/what-are-the-key-hyperparameters-for-a-random-forest-model/>)

Unsupervised Learning

69. What is Unsupervised learning, and what are its main types?
(<https://aiml.com/what-is-unsupervised-learning/>)
70. What is Clustering in unsupervised learning? (<https://aiml.com/what-is-clustering/>)
71. What are some common clustering algorithms, and how do they work?
(<https://aiml.com/what-are-the-most-common-categories-of-clustering/>)
72. How does dimensionality reduction help in unsupervised learning?
(<https://aiml.com/what-is-dimensionality-reduction/>)
73. Explain the difference between principal component analysis (PCA) and t-SNE (<https://aiml.com/how-does-t-sne-compare-to-pca/>)
74. What is Principal Component Analysis (PCA), and how does it differ from clustering? (<https://aiml.com/what-is-principal-component-analysis-pca->

and-how-does-it-differ-from-clustering/)

75. How do you evaluate the quality of clustering results in unsupervised learning? (<https://aiml.com/what-are-some-common-evaluation-metrics-in-clustering/>)
76. How does K-means work? What are some pros and cons of K-Means Clustering? (<https://aiml.com/how-does-k-means-work/>)
77. What are some common distance metrics that can be used in clustering? (<https://aiml.com/what-are-some-common-distance-metrics-that-can-be-used-in-clustering/>)

Data Preprocessing and Feature Engineering

78. What is Feature Scaling? Explain the different feature scaling techniques (<https://aiml.com/what-is-feature-scaling-explain-the-different-feature-scaling-techniques/>)
79. What are some common Feature Engineering techniques? (<https://aiml.com/what-are-some-common-feature-engineering-techniques/>)
80. How are Categorical Features represented? (Explain both one-hot and ordinal encoding) (<https://aiml.com/how-are-categorical-features-represented-explain-both-one-hot-and-ordinal-encoding/>)
81. What is the curse of dimensionality, and how does it affect machine learning models? (<https://aiml.com/what-is-the-curse-of-dimensionality/>)
82. How can you deal with outliers in your data? (<https://aiml.com/what-are-some-automatic-outlier-detection-mechanisms/>)
83. What is the difference between Feature Engineering and Feature Selection? (<https://aiml.com/what-is-the-difference-between-feature-engineering-and-feature-selection/>)
84. What is Feature Standardization (or Z-Score Normalization), and why is it needed? (<https://aiml.com/what-is-feature-standardization-and-why-is-it-needed/>)

85. How do you handle missing data in a dataset? (<https://aiml.com/what-are-different-ways-to-impute-missing-values-for-a-feature/>)

Model Evaluation and Optimization

86. What is cross-validation, and why is it important in model evaluation? (<https://aiml.com/how-does-cross-validation-work/>)

87. How are model hyper-parameters generally selected? (<https://aiml.com/how-are-model-hyper-parameters-generally-selected/>)

88. What is the purpose of regularization in machine learning models? (<https://aiml.com/what-is-regularization/>)

89. What is the bias-variance tradeoff and how do you balance it? (<https://aiml.com/what-is-the-bias-variance-tradeoff/>)

90. What are learning curves, and how do they help in model assessment? (<https://aiml.com/how-does-a-learning-curve-give-insight-into-whether-the-model-is-under-or-over-fitting/>)

91. How does gradient descent work, and how is it used in training machine learning models? (<https://aiml.com/what-is-gradient-descent/>)

Statistics

92. What is a p-value, and what is its significance? (<https://aiml.com/what-is-a-p-value-and-what-is-its-significance/>)

93. Describe a confidence interval (<https://aiml.com/what-is-a-confidence-interval/>)

94. Explain Bayes' Theorem (<https://aiml.com/what-is-bayes-rule/>)

95. How would you conduct an A/B test?

96. What is the difference between parametric and non-parametric models? (<https://aiml.com/what-is-the-difference-between-parametric-and-non-parametric-models/>)

97. What is the difference between Mean, Median and Mode?
(<https://aiml.com/differentiate-between-mean-median-and-mode/>) How to choose between mean and median to summarize data?
(<https://aiml.com/when-is-median-a-better-measure-of-a-distribution-than-mean/>)
98. How does Bayesian Statistics differ from the Frequentist paradigm?
(<https://aiml.com/how-does-bayesian-statistics-differ-from-the-classical-paradigm/>)
99. What is the Central Limit Theorem (CLT), and what are its implications for statistical inference? (<https://aiml.com/what-is-the-central-limit-theorem-clt-and-what-are-its-implications-for-statistical-inference/>)
100. What is Skewness and Kurtosis? (<https://aiml.com/what-is-skewness-and-kurtosis/>)
-

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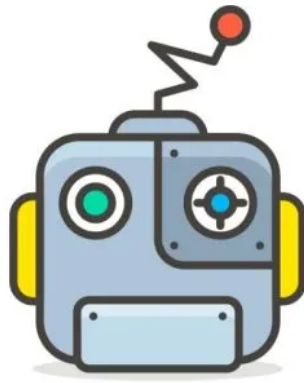
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